Perfect 👍 — you've uploaded two lecture slide decks:

1. **Prologue – The Software Process**
2. **Programming Review and Introduction to Software Design**

You want each slide and topic explained in an **easy-to-understand way (80% focus on understanding, 20% detail/technical)** — basically like how a teacher would explain the concepts clearly.

Let's start with **Lecture 1: "Prologue – The Software Process."**

🧩 Lecture 1: The Software Process (Overview + Explanation)

1. Software Process:

A **software process** is simply the **step-by-step method** followed by developers to create a working software product.

🧠 Easy Understanding:

Think of it like a **recipe** — if you're baking a cake:

- You first gather ingredients (requirements)
- Then decide how to bake it (design)
- Actually bake (coding)
- Taste test (testing)
- And fix or improve it later (maintenance)

⚙️ Main Phases:

Phase	Purpose	Question it Answers
Requirements Analysis	Define what the software must do	"What should it do?"
Design	Decide how it will do it	"How should it be built?"
Implementation (Coding)	Write actual code	—
Testing	Check for errors	—

Phase	Purpose	Question it Answers
Maintenance	Fix bugs & add new features	—

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2. Personal Finance Example

This example shows all stages applied to a simple “bank account” program:

- **Requirement:** The app must show user’s balance.
- **Design:** Create classes like CheckingAccount, SavingsAccount.
- **Implementation:** Write the actual Java/Python code.
- **Testing:** Run deposit/withdraw cases to check correctness.
- **Maintenance:** Fix issues or add new features (like new currency).

🧠 *Think of it as a “mini project” explaining the whole process.*

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3. Waterfall Process

A **linear process model** — each step flows into the next like a waterfall.

1. Requirements →
2. Design →
3. Coding →
4. Testing →
5. Maintenance

🧠 **Good for small, clear projects.**
❌ **Not good when requirements change** — because once you move ahead, it’s hard to go back.

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4. Why Pure Waterfall is Not Practical

- Clients rarely know everything at the start.
- We often learn and refine requirements during design or coding.

- Developers can't sit idle waiting for one stage to finish.
- Real-world projects often have **overlapping phases.**

🧩 *So, in practice — teams mix stages or use iterative models (like Spiral or Agile).*

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5. Spiral Process

This is an **iterative** model — you keep looping through:

1. Plan
2. Design
3. Build
4. Test
5. Evaluate
   — until the final product is complete.

🌀 *It's like improving your cake recipe each time you bake it.*

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6. Common Procedures

Two important habits in every phase:

- **Work against the prior phase:** Each stage builds on the previous one (you code what was designed).
- **Inspections:** Carefully reviewing documents or code to catch mistakes early — before they grow bigger.

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7. Requirements Analysis

Understand **what** users need.
🧠 Key idea: "Don't assume — clarify."

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8. Challenges of Requirements

Writing requirements clearly and updating them is hard.
They must be:

- In simple, non-technical English
- From **user's perspective**
- Organized and easy to update

Example: "App shall allow user to reset password via email."

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9. Design – The Heart of Software

Design answers **"how to build it."**
It shows:

- The components (classes, functions)
- How they connect (architecture)
- How data flows

It's like an engineer's blueprint before construction.

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10. Coding Tips

- Only code from a **well-defined design**
- Clearly state what each function/method does
- Review your code before compiling ("author inspection")

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11. Build-a-little, Test-a-little

Instead of writing the whole program and then testing —
✅ Write small parts → Test immediately → Fix → Move on

🧩 This reduces bugs and improves confidence.

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12. Testing

Goal: Check if software meets the requirements.

🧠 Important:
Testing **finds defects**, it **never proves** there are *no* defects.

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13. Types of Testing

Type	Purpose
Black-box testing	Test input/output — ignore internal code
White-box testing	Test logic, branches, and loops
Unit Testing	Test smallest parts (functions/classes)
Integration Testing	Test how parts work together
System Testing	Test the whole application

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14. Testing Tips

- Test **early** and **frequently**
- Include **extreme or invalid** values
- Change test cases often

🧠 Example: Try entering 0, 999999, or letters in a numeric field.

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15. Maintenance

Work after software delivery:

Type	Meaning
Corrective	Fix defects
Perfective	Add new features
Adaptive	Adapt to new systems/environments
Preventive	Make changes to prevent future failures

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16. Summary

The **software process** is a systematic way of creating and maintaining software:

- Define → Design → Code → Test → Maintain
- Based on models like **Waterfall**, **Spiral**, or **Agile**.

Would you like me to move on to **Lecture 2: "Programming Review and Introduction to Software Design"** next — in the same **80/20 understanding-based style**?

Perfect ✅ — let's continue with **Lecture 2: "Programming Review and Introduction to Software Design."**
I'll explain **each section** (like a teacher) with **80% focus on understanding** and only **20% technical detail**, so it feels clear and practical.

🧠 Lecture 2: Programming Review and Introduction to Software Design

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1. Purpose of the Lecture

This lecture helps you:

- Review good programming habits
- Learn **how to document and describe functions** properly
- Understand **what software design is**
- Know the **goals and qualities** of a good software design

Think of it as:

"How to write clean, understandable, and well-planned code before you even start coding."

🧩 Part 1 — Documenting Functions

2. What is Function Documentation?

When you write a function or method, you must describe:

- What it **expects** before running

- What it **does**
- What it **returns** afterward
- What issues or limits exist

This helps **any other developer (or future you)** understand the code easily.

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3. Preconditions, Postconditions, and Invariants

These are the **rules** that define a function's behavior.

Term	Meaning	Example
Precondition	What must be true before the function runs	open == true before using add()
Postcondition	What must be true after it runs	balance' == balance + anAmount
Invariant	What never changes during execution	"balance must always be between -1000 and 1000"

🧠 Simple view:
Pre = Before condition ✅
Post = After condition ✅
Invariant = Always true 🔁

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4. Example Exercise

Suppose you have a class **Account** with:

- boolean open
- int balance (must stay between -1000 and +1000)
- Method add(int anAmount)

You must document this method.

✅ **Documentation Example:**

Preconditions: open == true

-1000 <= balance <= 1000

-1000 <= balance + anAmount <= 1000

Postcondition: balance' == balance + anAmount

Returns: balance'

🧠 This shows the method's **contract** — what it promises to do and under what conditions.

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5. Describing How a Function Works

You can describe a function using:

- **Activity Diagrams (Flowcharts)** — visual representation
- **Pseudocode** — step-by-step logic in plain English

📘 Example:

"If the balance is valid, add the amount; else show an error."

🧠 These methods help visualize algorithms **before coding**, saving time and errors later.

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6. Example: Address Book Flowchart

A flowchart might show how a simple address book program works:

- Ask user → "Add" or "Retrieve"?
- If Add → Enter name & address → Save to file
- If Retrieve → Ask name → Show address

✅ The diagram helps you (and others) understand the logic quickly.

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7. Pseudocode Example: X-ray Controller

This example shows how pseudocode expresses logic **without syntax**:

FOR number of microseconds supplied by operator

  IF exceeds critical value

    try to get supervisor approval

    IF no approval → abort

ENDIF

IF power level too high → abort

IF patient aligned & shield placed & self-test passed

Apply X-ray

ENDFOR

🧠 *It's readable like English — no programming syntax required.*

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8. Pros & Cons of Pseudocode and Flowcharts

✅ **Advantages**

- Clarifies algorithms
- Catches design errors early
- Increases reliability
- Easier communication between team members

❌ **Disadvantages**

- Extra documentation to maintain
- Errors can occur while converting to code
- Sometimes need special tools for diagrams

🧠 *But overall, these tools are worth using — especially in big projects.*

🧩 **Part 2 — Good Habits for Writing Functions**

9. Good Habits (1/2)

Habit	Meaning	Example
Expressive names	Function & variable names should show purpose	Bad: manipulate() → Good: getBaseRaisedToExponent()

Habit	Meaning	Example
Avoid global variables	Pass parameters instead of using shared globals	Makes debugging easier
Defend against bad data	Check inputs & handle errors gracefully	Use conditions or exceptions

🧠 Always write **self-explanatory** code — so it "tells a story" by itself.

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10. Good Habits (2/2)

- Give **names to numbers** instead of using raw values
- int NUM_CELLS = 8927;
- for (i = 0; i < NUM_CELLS; i++)
- Limit parameters (ideally ≤ 6–7)
- Initialize variables near first use
- Check loops for correctness and termination
- Avoid deep nesting (no more than 3 levels)
- Inspect your code before compiling

🧠 These are like "clean code commandments" — they make your code readable and bug-free.

🧩 **Part 3 — Software Design Basics**

11. What is Software Design?

A **software design** is a complete plan or blueprint that guides programmers to write the code.

🧠 Just like architects use blueprints for buildings, developers use **software design documents** to build programs.

It must be detailed enough that a developer can code directly from it — without confusion.

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12. Goals of Software Design

Design turns **requirements (WHAT)** into a **solution (HOW)**.

It describes:

- The **system structure** (architecture)
- How components **fit together**
- How each part can be coded

🧩 Essentially: *Design bridges the gap between user needs and coding.*

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13. Key Qualities (Goals) of a Good Design

Goal	Meaning
Correctness	Meets all requirements
Robustness	Handles errors gracefully
Flexibility	Easy to modify or expand
Reusability	Can be reused in other apps
Efficiency	Uses resources wisely
Reliability	Works consistently without failure
Usability	Easy for users to operate

🧠 These are the "7 pillars" of good design.

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14. Explanation of Each Goal

🟢 **Correctness**

The software must do exactly what was required — no more, no less.

🟢 **Robustness**

It should survive unexpected conditions (invalid input, user mistakes, etc.)
Example: Program shouldn't crash if user enters "abc" instead of a number.

🟢 **Flexibility**

Easy to change or extend later.
Example: Adding a new feature like “withdraw” in a banking app shouldn’t break the system.

🟢 **Reusability**

Write components that can be reused elsewhere.
Example: The same Customer class used in both billing and support systems.

🟢 **Efficiency**

Use CPU and memory wisely — fast and lightweight software is always better.

🟢 **Reliability**

It should rarely fail.
Measured by “average time between failures.”

🟢 **Usability**

Users should find it simple and intuitive to use.

🧩 **Part 4 — Example: Command Line Calculator**

15. Requirements

- Ask how many accounts to open (start with zero balance).
- Let user choose which account to use.
- Allow adding/subtracting money.
- User can switch accounts or quit.

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16. Implementation Issues

Problems often appear like:

- Invalid input → crash
- Wrong account number → crash
- Invalid string commands

🧠 These show poor **robustness** and **flexibility.**

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17. Design Improvement

You can improve by applying design goals:

- Add **error checking** (robustness)
- Make it easy to **add features** (flexibility)
- Make code **modular** and **reusable**
- Ensure it runs fast (efficiency)

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18. Summary

A **software design** is the complete blueprint for building software.

🎯 **Goals Recap:**

- Correctness
- Robustness
- Flexibility
- Reusability
- Efficiency
- Reliability
- Usability

🧠 In short:

"Good design means building software that works correctly, handles mistakes, adapts to change, and is easy to maintain."

Would you like me to make a **summary sheet / short revision notes (2 pages)** for both lectures combined — ideal for exam revision (concepts + examples)?

Perfect ✅ — now that you've uploaded all **three remaining lectures**, here's the plan:
I'll explain each —

- **Lecture 3: Object Orientation**

- **Lecture 4: Design Principles I (Correctness & Robustness)**
- **Lecture 5: Design Principles II (Flexibility, Reusability & Efficiency)**
  in the **same clear, 80/20 understanding-based style** (like a teacher summarizing + explaining the real meaning of slides).

🎯 Lecture 3 — Object Orientation

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1. What is Object Orientation?

Object Orientation (OO) is about **thinking in terms of real-world objects** — turning real-world things (like Customer, Account, Car) into **software objects** that contain:

- **Data** (attributes)
- **Behavior** (methods/functions)

🧠 Example:
In a bank —
Customer, Account, and Transaction are all *objects* that interact.

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2. Goal of Object Orientation

OO helps us describe software like we describe the **real world**:

"Customers Montague and Susan entered the bank and were served by teller Andy."

Here:

- **Customer** → Class
- **Montague, Susan** → Objects
- **Teller** → Another object interacting with customers

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3. Real-World Mapping

Every object in software should **represent something real**.
Example:

Real-world Concept	Software Representation
Customer	class Customer {}
Account	class Account {}
Transaction	class Transaction {}

This **direct mapping** makes software easier to understand and maintain.

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4. Benefits of Object Orientation

- ✅ Easier to design complex systems
- ✅ Encourages code reuse
- ✅ Easier maintenance
- ✅ Models real-world relationships naturally

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5. Classes & Objects

Term	Meaning
Class	Blueprint or template describing what an object is and what it can do.
Object	Actual instance of a class (a real example).

🧠 Example:
Class → Car
Object → myCar, toyotaCorolla2020

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6. Members of a Class

Each class has:

- **Attributes (data):** e.g., mileage, vehicleID
- **Methods (behavior):** e.g., start(), stop()

Subclasses inherit these members too.

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7. Types of Attributes

Type	Meaning
Naming	Unique for each object (e.g., vehicleID)
Descriptive	Changes during object's life (e.g., mileage)
Referential	Connects one class to another (e.g., owner of car)

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8. Clients of a Class

A **client class** uses another class's methods.

Example:

```
class Customer { int computeBalance() { ... } }


class Bank {
  void showAssets() {
    Customer c = new Customer();
    int balance = c.computeBalance();
  }
}
```

Here, Bank is a **client** of Customer.

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9. Why OO is Useful

Object orientation supports:

- **Abstraction:** focus on essential details only
- **Encapsulation:** hiding data within objects
- **Inheritance:** reuse through hierarchy
- **Polymorphism:** same action, different results depending on context

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10. Example – Email Creation System

Different types of customers need different email messages (delinquent, regular, mountain).
OO allows you to create:

- Base class → Customer
- Subclasses → DelinquentCustomer, RegularCustomer, etc.

Each subclass defines its own message format — no messy if-else chains.

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11. Problem with Non-OO Code

Using many if or switch conditions makes code:

- Hard to read
- Difficult to modify
- Expensive to maintain

OO solves this using **polymorphism**.

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12. Polymorphism

Means **"many forms."**
The same function name behaves differently for each subclass.

Example:

Customer c = new RegularCustomer();

c.sendEmail(); // sends regular promo

c = new DelinquentCustomer();

c.sendEmail(); // sends warning email

---

13. Interfaces

Interfaces group related methods and provide **clear contracts** for what a class should do.
Example:

```
interface Drawable {
  void setColor(int color);
  void draw();
}
```

They help **organize complex classes** (like separating Pen, Color, Logo behaviors).

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14. Summary

- **Class:** defines a concept
- **Object:** an instance of the class
- **Inheritance:** "is-a" relationship
- **Aggregation:** "has-a" relationship
- **Polymorphism:** same method, different behavior

🧠 OO makes programs realistic, maintainable, and flexible.

🎯 Lecture 4 — Design Principles I: Correctness & Robustness

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1. Design Principles Overview

Focuses on **two key principles**:

- **Correctness** – does it do what it's supposed to?
- **Robustness** – can it handle errors gracefully?

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2. Correctness

A correct design **meets all requirements** exactly as specified.

Two ways to ensure correctness:

1. **Informal approaches** – logical reasoning, simplification, modular design
2. **Formal approaches** – mathematical logic (rarely used outside critical systems)

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3. Informal Correctness

Make designs:

- **Readable** (so they're understandable)
- **Modular** (divide complex system into small parts)

🧠 "If you can't understand your design, you can't claim it's correct."

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4. Formal Correctness

Uses **invariants** — rules that remain true throughout program execution.

Example (Car class):

mileage > 0

mileage < 1,000,000

value >= -300

(type == "REGULAR" && value <= originalPrice) ||

(type == "VINTAGE" && value >= originalPrice)

These conditions define *correct states* of the object.

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5. Using Invariants in Code

Keep variables **private** and change them only through **accessor methods** that preserve invariants.

🧠 Example: Setters that reject invalid data.

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6. Interfaces and Modularity

- **Interface:** defines available functions (method signatures)

- **Modularity:** breaking design into clear, separate parts (classes/packages)

Interfaces make complex designs **manageable and testable**.

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7. Domain vs. Non-Domain Classes

Type	Meaning	Example
Domain Classes	Specific to your application	Customer, Transaction
Non-Domain Classes	Generic helpers	Logger, DatabaseConnector

A good design uses both to keep code organized and reusable.

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8. Robustness

The ability to handle unusual or wrong conditions **without crashing**.

Sources of errors:

- Invalid user input
- Data communication errors
- Developer mistakes

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9. How to Improve Robustness

✅ Check input validity before using it
✅ Verify types and ranges
✅ Initialize variables properly
✅ Use exceptions or default values

Example:

if (aLength <= 0) throw new IllegalArgumentException("Invalid length");

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10. Wrapping Parameters

Instead of passing raw data:

computeArea(int a, int b);

Use a class:

computeArea(Rectangle rect);

This ensures parameters are always valid (robustness + correctness).

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11. Design Detail Balance

Too much design = slow progress
Too little = confusion later

🧠 Rule: Simple tasks need little design, complex systems need full design documentation.

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12. Summary

Principle	Meaning
Correctness	Design must meet all requirements
Robustness	Design must handle unexpected conditions gracefully

Together, they make your system **reliable and trustworthy**.

🎯 Lecture 5 — Design Principles II: Flexibility, Reusability & Efficiency

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1. Flexibility

Ability to **easily change or extend** software.

🧠 Example: In a banking system, adding a new type of account shouldn't break existing code.

Ways to achieve flexibility:

- Use **abstract classes** or **interfaces**
- Use **inheritance** for future extensions
- Avoid "hard-coding" things that may change

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2. Reusability

The ability to use existing code **in new contexts** without modification.

Tips:

✅ Write methods independent of specific classes (use static if possible)
✅ Keep methods well-documented and parameterized
✅ Use expressive, meaningful names
✅ Reduce dependencies between classes

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3. Improving Class Reusability

- Make class represent a real-world concept
- Reduce unnecessary links to other classes
- Keep it general enough to fit future needs

🧠 Example: Instead of OnlineStudent, create Student and reuse for online or campus.

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4. Efficiency

Means **using time and memory wisely** — without sacrificing flexibility.

Two dimensions:

- **Time efficiency:** how fast it runs
- **Space efficiency:** how much memory it uses

🧠 Always balance efficiency with readability and maintainability.

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5. Trade-offs

Sometimes you must **compromise**:

Design Goal	Possible Trade-off
More flexibility	May reduce speed
More reusability	May require extra memory

Design Goal	Possible Trade-off
More efficiency	May reduce clarity

---

6. Practical Techniques

- Use **profilers** to detect performance bottlenecks
- Reduce **loop complexity**
- Minimize **network calls**
- Store only essential data

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7. Example – Calculator Design

Flexible design:

- Separate logic into **classes (Add, Multiply, Divide)**
- Each operation extends base Calculator class
  → Easy to add new operations later.

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8. Summary

Principle	Meaning
Flexibility	Easy to change or extend
Reusability	Useful across multiple projects
Efficiency	Performs well in time and space

🎯 **Good design balances all three** — flexible enough to change, reusable for other work, and efficient enough to perform well.

Would you like me to make a **combined "Final Revision Notes" file (PDF or DOCX)** — with summarized points, examples, and diagrams from **all five lectures** (Prologue → Design Principles II) in one easy-to-study format?